

BILASH DEVNATH

PhD Candidate, Chemical Engineering | Resource Recovery • Separations • Sustainable Process Engineering

RESEARCH PROFILE

Chemical Engineering PhD candidate with research spanning nutrient and critical-material recovery, hydrothermal and thermochemical processing, solid-liquid separations, sorbent/material modification, and process-scale assessment. Develops experimental workflows from reaction and selective mobilization through capture, product characterization, and performance validation, with additional experience in techno-economic analysis (TEA), life-cycle assessment (LCA), and multidisciplinary materials research.

EDUCATION

PhD Candidate in Chemical Engineering, Florida Institute of Technology, Melbourne, FL
GPA: 4.00/4.00

Aug 2023 - Present

Bachelor of Science in Chemical Engineering, Bangladesh University of Engineering and Technology (BUET), Dhaka
GPA: 3.73/4.00

Apr 2018 - May 2023

RESEARCH & PROCESS DEVELOPMENT EXPERIENCE

Graduate Researcher | Reza Research Group, Florida Institute of Technology

Aug 2023 - Present

- Develop and execute hydrothermal and chemical processes for resource recovery from animal manure, agricultural biomass, and carbon-based materials using high-temperature/high-pressure reactors, acid/base treatment, filtration, precipitation, solvent extraction, and post-treatment.
- Developed an acid-mediated mild hydrothermal process for phosphorus recovery from dairy manure; 87.6-91.5% of manure phosphorus was mobilized to the aqueous phase, followed by 87.7-91.7% recovery of dissolved phosphorus through alkaline precipitation into a slow-release multi-nutrient matrix.
- Connect reaction conditions with product yield, elemental composition, nutrient partitioning, surface chemistry, porosity, thermal stability, agronomic behavior, and environmental performance.
- Characterize solid and liquid products using ICP-based elemental analysis, ion chromatography, FTIR, TGA, CHNS, LC-MS, SEM/EDS, XRD, BET surface-area/pore analysis, UV-Vis spectroscopy, and water-contact-angle analysis.
- Develop process-scale TEA and LCA models incorporating mass and energy balances, equipment requirements, utilities, sensitivity analysis, economic metrics, and environmental tradeoffs.
- Contribute to collaborative critical-material and functional-material research, including rare-earth adsorption using magnetically retrievable core@shell nanocomposites and selective metal mobilization from waste printed circuit boards.
- Prepare first-author manuscripts, conference presentations, technical documentation, and publication-quality analyses; mentor undergraduate researchers and support shared-instrument operation and troubleshooting.

In-Plant Trainee | Eastern Refinery Ltd., Bangladesh

Apr 2022

- Observed atmospheric distillation, catalytic reforming, large-scale separation, process control, and industrial waste-management operations in a petroleum refinery environment.

TECHNICAL EXPERTISE

Process & separations: Hydrothermal conversion; pyrolysis; acid/base extraction and leaching; precipitation; filtration; solid-liquid separation; solvent extraction; adsorption/recovery studies; drying; chemical post-treatment; experimental design; mass and nutrient balances.

Characterization: FTIR; TGA; CHNS; UV-Vis; LC-MS; ion chromatography; SEM/EDS; XRD; BET; ICP-based elemental analysis; water contact angle.

Modeling & data: Aspen HYSYS; COMSOL Multiphysics; MATLAB; R; Python; Excel-based process modeling; TEA; LCA; sensitivity analysis; statistical data analysis; AutoCAD; Visio; Adobe Illustrator.

PEER-REVIEWED PUBLICATIONS & MANUSCRIPTS

- Devnath, B.; Gao, B.; Reza, M. T. Recovery of slow-release multi-nutrient matrix from dairy manure through acid-mediated mild hydrothermal treatment. *Waste Management* 2026, 224, 115732. doi:10.1016/j.wasman.2026.115732
- Devnath, B.; Khanal, S.; Shah, A.; Reza, M. T. Nutrient recovery from co-hydrothermal carbonization of animal manures: Synergistic effects on hydrochar properties and agronomic potential. *Waste Management* 2026, 213, 115391. doi:10.1016/j.wasman.2026.115391

- Devnath, B.; Khanal, S.; Shah, A.; Reza, M. T. Hydrochar and biochar from dairy manure as stable carbon amendments for greenhouse gas mitigation and carbon sequestration in concentrated animal feeding operations. *Environmental Research Communications* 2026, 8, 065015. doi:10.1088/2515-7620/ae74a0
- Devnath, B.; Khanal, S.; Shah, A.; Reza, M. T. Influence of hydrothermal carbonization (HTC) temperature on hydrochar and process liquid for poultry, swine, and dairy manure. *Environments* 2024, 11, 150. doi:10.3390/environments11070150
- Shokouhimehr, M.; Guidugli, L. F.; Smith, R. C.; Knight, C.; Devnath, B.; Reza, M. T. Magnetically retrievable core@shell nanocomposites for rare earth element adsorption: Experimental and machine learning insights. *Langmuir* 2026, 42(28), 20456-20467. doi:10.1021/acs.langmuir.6c02166
- Bevers, N.; Timilsina, A. P.; Manandhar, A.; Ansari, A.; Klingman, M.; Witter, J.; Arnold, G.; Shah, A.; Devnath, B.; Reza, M. T.; Khanal, S. Evaluating the impact of manure-derived hydrochar soil amendments on greenhouse gas fluxes: A greenhouse experiment using soil mesocosms and corn plants. *Journal of Environmental Management* 2026, 414, 130447. doi:10.1016/j.jenvman.2026.130447
- Olushola, S. B.; Bocus, K. M. A.; Devnath, B.; Reza, M. T.; Pahinkar, D. G. Durable, hydrophobic and porous zeolite 13X coatings for resisting liquid water during CO₂ desorption. *Surfaces and Interfaces* 2026, 83, 108554. doi:10.1016/j.surfin.2026.108554
- Guidugli, L. F.; Devnath, B.; Reza, M. T. Hydrothermal carbonization process liquids as hydrogen bond donors for hydrophobic deep eutectic solvents: COSMO-RS design, physicochemical characterization, and methylene blue extraction. *Manuscript under review*

SELECTED CURRENT RESEARCH DIRECTIONS

- Sequential nutrient-release profiling of recovered fertilizer products under water, pH-responsive, rhizosphere-mobilizing, and residual extraction conditions.
- Scale-up, TEA, and LCA of a centralized hydrothermal dairy-manure treatment and nutrient-recovery facility.
- Hydrothermal activation and modification of biochar to improve nutrient adsorption and retention behavior.
- Selective metal mobilization and staged critical-material recovery from waste printed circuit boards.
- Hydrochar post-treatment, phytotoxicity/ecotoxicity reduction, and fire-safety assessment for handling and agricultural deployment.
- ML-guided biochar/hydrochar nutrient retention and release using Python-based XGBoost/random-forest workflows and feature-importance analysis.

SELECTED CONFERENCE PRESENTATIONS

- Devnath, B.; Reza, T. Integrating hydrochar and biochar production into dairy CAFO manure management: A pathway to carbon sequestration and greenhouse gas mitigation. AICHE Annual Meeting, Boston, MA, 2025.
- Devnath, B.; Reza, T. Optimizing manure management in CAFOs: Hydrochar and biochar production for carbon sequestration. Florida Section of ASABE Annual Meeting, Jupiter Beach, FL, 2025. Outstanding Student Presentation.
- Devnath, B.; Reza, T. Enhancing carbon sequestration in soil: Comparative analysis of hydrochar and biochar. Florida Academy of Sciences Annual Meeting, Melbourne, FL, 2025. Outstanding Graduate Student Presentation.
- Devnath, B.; Reza, T. Effect of hydrothermal carbonization temperature and time on nutrient recovery from poultry litter. Hydrochar and Biochar Conference, Rapid City, SD, 2024. 2nd Best Student Presentation.
- Devnath, B.; Reza, T. Co-hydrothermal carbonization of corn stover and animal manure: Enhancement in fuel characteristics. ASABE Annual International Meeting, Anaheim, CA, 2024. Presentation Excellence Award.
- Devnath, B.; Reza, T. Hydrothermal carbonization of animal manure: Influence of temperature. ASABE Annual International Meeting, Anaheim, CA, 2024.
- Devnath, B.; Reza, T. Hydrothermal carbonization of animal manures: Effects of temperature on solid and liquid products. Florida Section of ASABE Annual Meeting, Jensen Beach, FL, 2024.

AWARDS & HONORS

- Outstanding Graduate Student of the Year, Florida Institute of Technology (2025-2026).
- Outstanding Graduate Student Presentation, Florida Academy of Sciences Annual Meeting (2025).
- Outstanding Student Presentation, ASABE Florida Section Annual Meeting (2025).
- NASA Florida Space Grant Consortium Award, Near Space Investigation Program, Center for Space Education (2025).
- Inducted into Phi Kappa Phi Honor Society, Florida Institute of Technology (2025).
- Presentation Excellence Award, ASABE Annual International Meeting (2024).
- 2nd Best Student Presentation, Hydrochar and Biochar Conference (2024).

TEACHING, MENTORING, LEADERSHIP & SERVICE

Teaching Assistant | Chemical Engineering / Materials

Fall 2023 - Spring 2024

Laboratory, Florida Institute of Technology

- Supported instruction in materials properties, experimental methods, laboratory practice, and data analysis.
- Mentor undergraduate researchers in experimental work, characterization, data interpretation, and technical communication.

President | Bangladeshi Student Association, Florida

2024 - 2026

Institute of Technology

- Led student-community programming, coordination, and organizational activities; supported committee transition and mentoring of new student leaders.

Vice President | Badminton Club, Florida Institute of

2026 - Present

Technology

- Support organization of practices and matches for a diverse campus community.

Professional memberships: American Institute of Chemical Engineers (AIChE); American Society for Agricultural and Biological Engineers (ASABE).